



UNIVERSITY OF LONDON

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**BUSINESS ANALYTICS, APPLIED MODELING & PREDICTION
COMMERCIAL INSIGHTS WITH TABLEAU**

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Executive Summary

This report provides a comprehensive analysis of the business's sales and profitability across different segments and markets from 2019 to 2022, leveraging data-driven insights derived from Tableau visualisations. Additionally, it will also identify the weakness and potentials of the company which help pinpoint the areas for improvement and growth of the business. The report consists mainly of five analyses: Sales and Profit Analysis, Product Performance Analysis, Customer Sales Insights, Geographical Analysis, Forecast and What-If Analysis.

The overall analysis highlights the overall positive performance of the business suggesting strong customer satisfaction and effective business operation. As for product performance, technology products are identified as the highest contributor for the sales. The consumer segment dominates sales among the customer segments. Geographical analysis identifies APAC as the foremost emerging market.

Besides, the analysis has also revealed certain weak points in the business which can be improved by the following recommendation:

1. Refine discount strategies for underperforming products.
2. Optimise marketing efforts to attract new customers.
3. Target new markets for expansion, with a focus on high-potential markets like APAC and Canada.
4. Implement additional measures to support regions facing challenging market conditions.

By implementing these recommendations, the business can drive sustainable growth, improve profitability, and maintain competitiveness in the global marketplace.

Introduction

In the ever-evolving realm of modern business, companies encounter unprecedented challenges and possibilities driven by technological advancements and global market dynamics. In order to maintain competitiveness, companies must constantly innovate and optimise their strategies.

This report aims to address these challenges by analysing the business data to identify and gain commercial insights. These insights are then used to help the business to make informed decisions and optimise business operations in order to better achieve the business objectives.

Tableau, a leading visualisation tool will be used to analyse the business data to obtain actionable insights. These insights will be presented in the form of a tableau story that consists of five dashboards: Sales and Profit Analysis, Product Performance Analysis, Customer Sales Insights, Geographical Analysis and Forecast and What-if Analysis.

Analysis

Sales and Profit Analysis

Overall Business Performance Review and Analysis

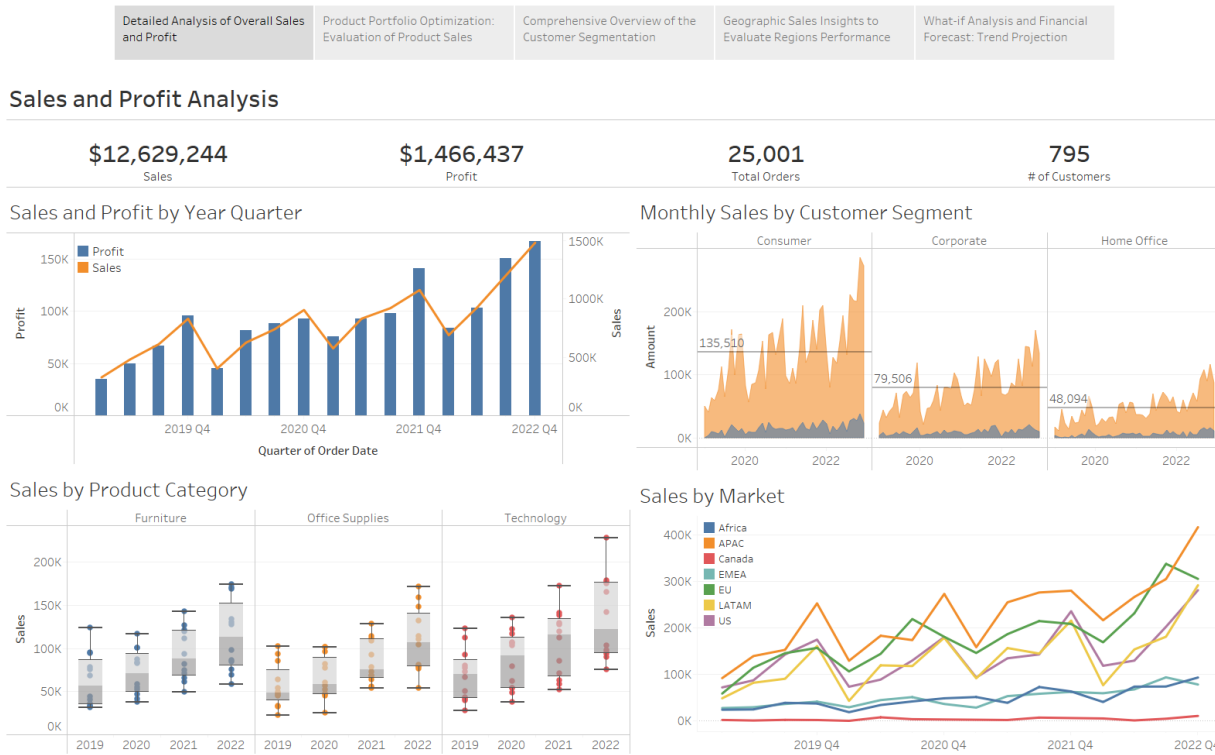


Figure 1: Dashboard for Sales and Profit Analysis

As shown in the dashboard, the total sales and profit from 2019 to 2022 were \$12,629,244 and \$1,466,437 respectively.

Based on the Sales and Profit by Year Quarter, the sales performance is promising, showing an increasing trend from 2019 to 2022. However, it is also observed that there is a seasonal sales trend indicating a consistent decrease every first quarter.

The Monthly Sales by Customer Segment shows an overall positive growth in sales for all 3 customer segments with consumers being the top contributor across the customer segments. This suggests that the business model is more aligned with the needs and preferences of individual consumers.

From the Sales by Product Category chart, it is identified that there is also an overall increasing trend for all the 3 product categories with technology products being the leading category in sales. The product category with the highest sales by month is technology products of \$227,423 sales in December 2022.

The Sales by Market chart shows the APAC market has the top sales among the markets. It is also noticed that there is a sharp increase in sales for APAC, LATAM, US in the 4th quarter of 2022 while a slight decrease in sales for EU and EMEA.

Based on dashboard insights, the business demonstrates overall positive performance. However, it is recommended to invest in off-season marketing to adapt to the seasonal sales trend to offset the seasonal decline and improve overall business profitability.

Product Performance Analysis

Overall Business Performance Review and Analysis

Detailed Analysis of Overall Sales and Profit	Product Portfolio Optimization: Evaluation of Product Sales	Comprehensive Overview of the Customer Segmentation	Geographic Sales Insights to Evaluate Regions Performance	What-if Analysis and Financial Forecast: Trend Projection
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Product Performance Analysis

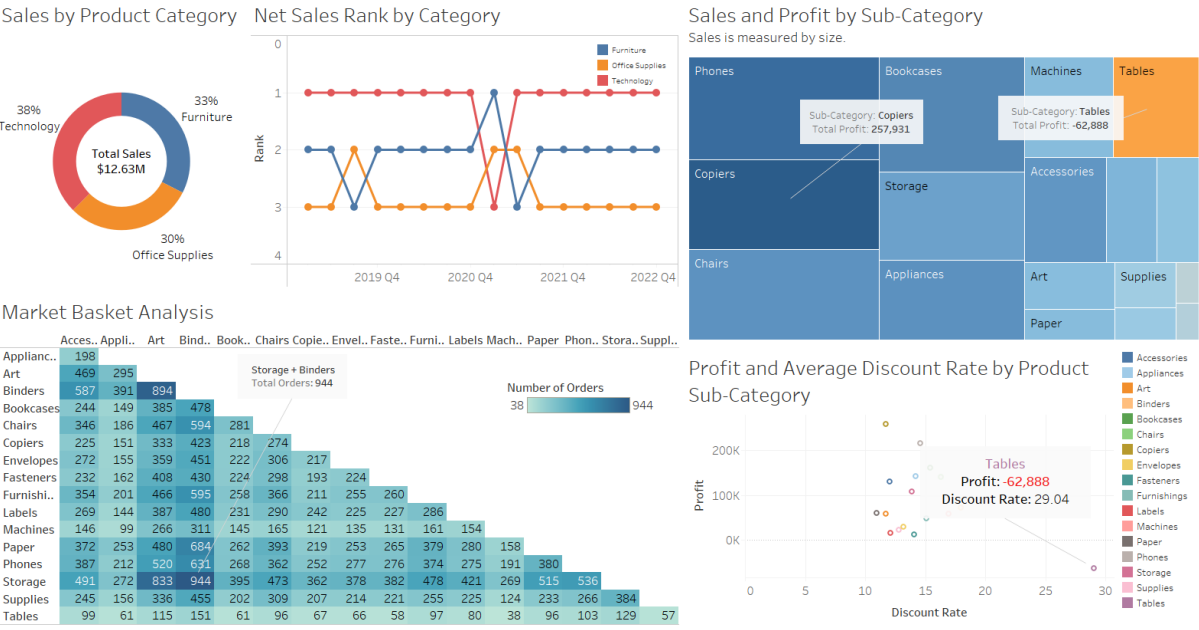


Figure 2: Dashboard for Product Performance Analysis

The business offers a diverse range of products in furniture, office supplies, and technology categories. Among the three categories, the highest sales contributor is the technological products, accounting for 38% of the total sales as shown in the sales by product category chart.

The net sales rank by product category shows that technology has been consistently leading in sales except for the first quarter of 2021, where furniture and office supplies temporarily surpassed technology. However, in the second quarter of 2021, technology regains its top position, outpacing furniture and office supplies again.

Based on the sales and profit by product sub-category, it can be identified that the most profitable product is copiers with the highest profit of \$257,931. Conversely, table is identified as the only loss-making product with the loss of \$62,888, despite achieving decent sales of \$755,993.

The market basket analysis chart revealed that the most popular product combination is storage and binders with an order count of 944. This suggests frequent bundle purchases of these items by customers.

The profit and average discount rate chart by product sub-category indicate that tables, despite having the highest discount rate at 29.04%, incur losses. It could be due to the discount strategy on tables not effectively offsetting the associated costs, thereby resulting in losses.

Based on the analysis, it is recommended to focus on the technology product category due to the increasing trend and demand of technological products. Additionally, a thorough review of the table product category is crucial to investigate the reasons behind the loss incurred. The discount strategy for table products should also be reassessed to improve the profitability without compromising sales.

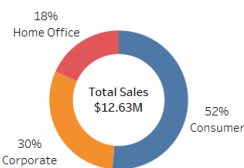
Customer Sales Insights

Overall Business Performance Review and Analysis

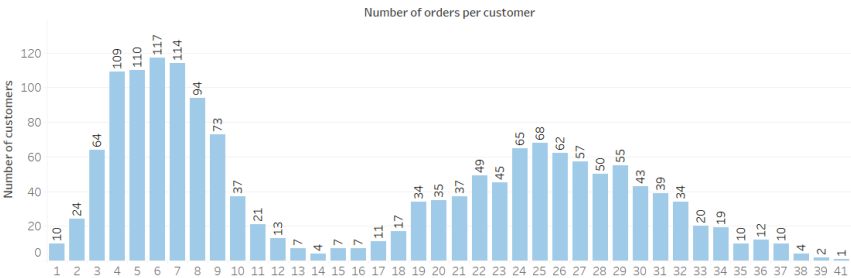
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Customer Sales Insights

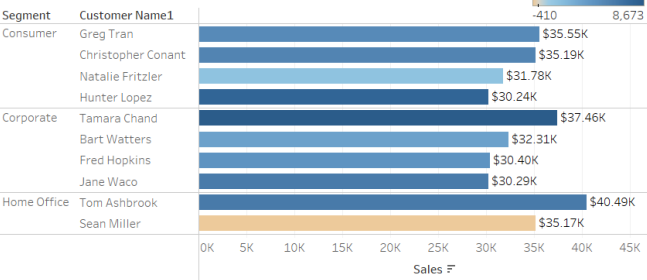
Sales by Customer Segment



Customer Loyalty



Top 10 Customers



Customer Cohort

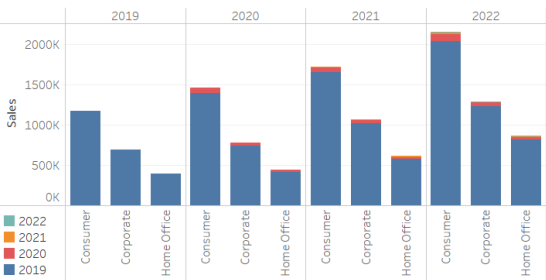


Figure 3: Dashboard for Customer Sales Insights

The customers were categorised into three segments: consumer, corporate and home office. Among these segments, the consumer segment emerges as the leading contributor, accounting for 52% of the total sales. In contrast, the home office segment accounts for the smallest portion of the total sales of 18%.

The customer loyalty chart shows that most of the customers make between 4 to 8 orders, with a smaller group making between 24-29 orders. Additionally, there is a notable customer who has demonstrated remarkable loyalty by making a total of 41 orders throughout the years.

Based on the Top 10 customers chart, Tom Ashbrook from the office segment stands out as the top sales contributor, contributing \$40.49k in sales to the business. Among the Top 10 customers, Sean Miller's sales contribution of \$35.17k with the accompanying loss of \$410 is notable.

Besides, the customer cohort chart groups the customers by the year they placed their first order and then analyse the sales from respective customer segments. It is noticed that the majority of sales are made by returning customers whose first order was made in 2019 and only a small portion of sales were made by new customers from 2020, 2021 and 2022.

Overall, the increasing sales and prevalence of returning customers demonstrate customer satisfaction with the products. However, optimising the marketing strategy to attract new customers is recommended to ensure continued growth and adapt into the dynamic marketplace.

Geographical Analysis

Overall Business Performance Review and Analysis

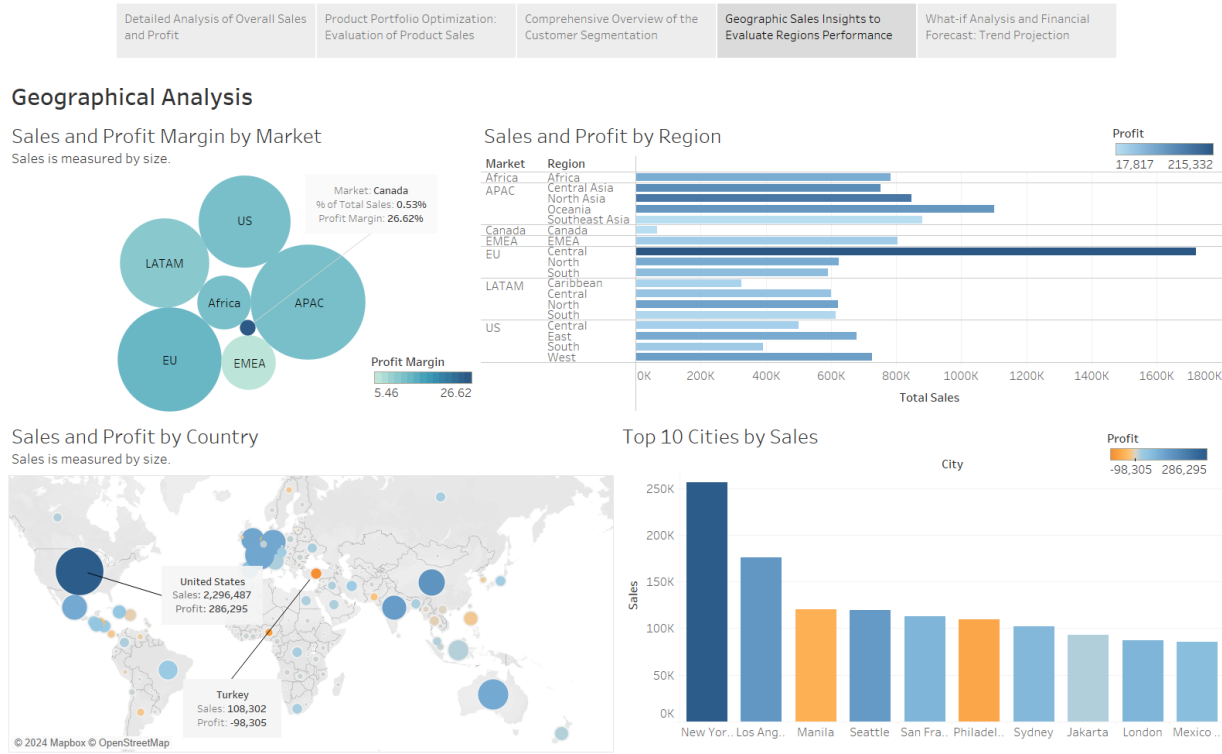


Figure 4: Dashboard for Geographical Analysis

The business operates globally, encompassing markets in US, LATAM, EU, APAC, EMEA, Canada and Africa. The sales and profit margin by market chart shows that Asia Pacific is the largest market for the business with the highest sales. The strong sales performance in the APAC market could be attributed to a larger customer base with greater consumer purchasing power, fuelled by rapid urbanisation. On the other hand, despite having the smallest market contributing only 0.53% of the total sales, Canada has the highest profit margin of 26.62% among the markets.

Upon examination of sales and profit by region within their respective markets, it becomes evident that the Central EU stands out with the highest sales of \$1,719,542 and highest profit of \$215,332. This suggests a strong market presence and effective business operations in the Central EU region.

The sales and profit by country map chart highlights the United States as the standout performer, with the highest sales of \$2,296,487 and the highest profit of \$286,295. However, it also reveals several

countries experiencing losses, with Turkey standing out as the highest loss-making country, reporting a loss of \$98,305.

From the top 10 cities sales and profit, New York emerges as the best-performing city with the highest sales of \$255,901 and the highest profit of \$62,002. However, it's notable that Manila and Philadelphia, despite having the top sales figures, both cities are experiencing losses.

Overall, several countries are experiencing losses, likely due to economic instability and operational challenges. To address this, it is recommended to implement additional business strategies to adapt to the challenging market conditions. Furthermore, expansion into the APAC market is recommended due to its emerging economic opportunities and exploration into the Canada market is advised, given its high profit margin showing its potential for substantial growth.

Forecast and What-If Analysis

Overall Business Performance Review and Analysis

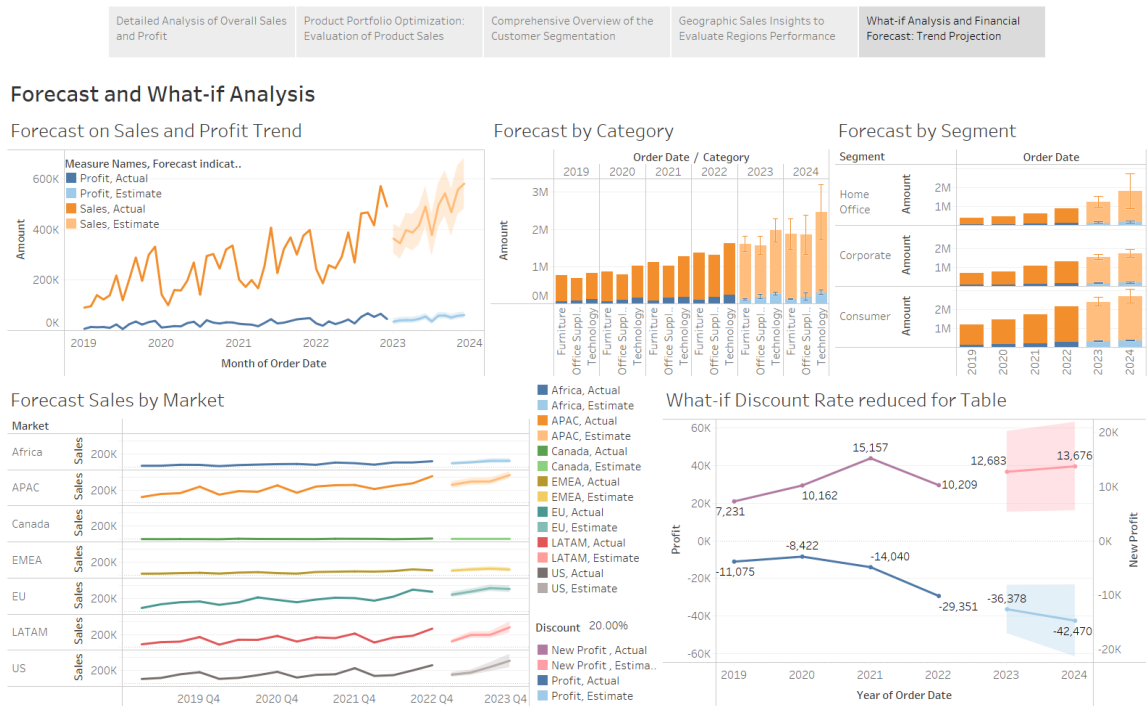


Figure 5: Dashboard for Forecast and What-if Analysis

The future forecast for sales and profit indicates that the increasing trend, following a similar pattern observed from 2019 to 2022, is expected to continue in 2023.

The sales forecast by product category indicates rising sales and profit projections for each category. Similarly, the forecast for sales by customer segment suggests a pattern of increasing sales and profit expected to continue in 2023. Likewise, the forecast for sales by market shows an overall uptrend in sales across all markets.

As mentioned in the product performance analysis, reassessing the discount strategy for table products is recommended. To visualise potential outcomes, a what-if analysis is performed to determine the impact of reducing the discount rate, assuming sales remain unaffected. As seen in the chart, if the average discount rate is reduced from 29.04% to 20.00%, the loss would be converted into profit, with an expectation of sustained profitability instead of further losses in the upcoming year.

Overall, the forecast implies that the business is expected to expand and grow in the foreseeable future.

Conclusion

In conclusion, the comprehensive analysis of sales and profit, product performance, customer insights, geographical sales, and future forecasts provides valuable insights for strategic decision-making.

The business exhibits promising growth potential in the global market, with technological products leading in sales and customer satisfaction evident. However, challenges such as seasonal sales trends and loss-making product categories require attention.

Recommendations include refining discount strategies, targeting new markets, and optimising marketing efforts. By implementing these recommendations and continuously monitoring their effectiveness, the business can optimise profitability and maintain competitiveness in the global marketplace while achieving sustainable growth.